

Corrections to *Who was done what? A parser-based study of passive voice constructions in media discourse on the Russo-Ukrainian War*

Natchanun Sanitdee

Version 1.1 · 3 October 2026

Helda: <http://hdl.handle.net/10138/585905> · Zenodo: [10.5281/zenodo.14031334](https://zenodo.org/record/14031334)

Before providing the details of the errors, I would like to add a personal note that my thesis was finished under time pressure during a serious illness in my family. Nothing has bothered me more than these loose ends I overlooked before submission and got hung in my head since.

Table C1. Summary of corrections.

#	Correction	Where in the thesis	Changes a finding?
1	The “t-score” is the z-score, and its p-values are not valid	§4.2.2, §5.1.1, §5.2.1, §5.2.2.1, §5.4.1	<i>Yes, two claims withdrawn</i>
2	The decline in passive verb use does not hold	§5.2.2.1, Table 18, Figure 8	<i>Yes, claim withdrawn</i>
3	Table 18’s %Pass. V. column uses two denominators	§5.2.2.1, Table 18	No
4	Table 25 contains a duplicated row	§5.3.1, Table 25	No
5	Russia/Russian as agent: the schema is computed on 35 of 37 clauses	§5.3.2, Table 29	No
6	“Leipzig 2014/2023” names two different preparations	§3.1.2, Tables 2 and 14	No
7	“Passive pair” is used for two different quantities	Tables 14 and 19	No
8	Table 4’s expected frequencies belong to another pair	§4.2.1–4.2.2, Table 4	No
9	Two Average rows are miscalculated	Tables 2 and 33	No
10	Abstract corpus size	Abstract	No
11	The links in Appendices 1 and 2 are dead	Appendices 1 and 2	No

WITHDRAWS A CLAIM

1. The “t-score” is the z-score, and its p-values are not valid (§4.2.2)

What the thesis says. §4.2.2 adopts the t-score as its association measure, defines it as $(O - E)/\sqrt{E}$, and supports the choice with Dennis (1965, cited in Evert 2004), Lijffijt et al. (2016) and Evert et al. (2008). Each score is converted to a p-value, and pairs with $p < 0.05$ and more than 2 occurrences (Ukraine War corpus) or 15 (Leipzig corpora) are reported as significant.

What is wrong. $(O - E)/\sqrt{E}$ is the z-score (Evert 2004); the t-score is $(O - E)/\sqrt{O}$. The two rank pairs differently, so the sources cited for the t-score do not support the measure actually used. The p-values come from a Student’s t distribution with one degree of freedom (`p_value_calculation_t-score.py`), which has no basis for either score, so they are not valid p-values. Under that conversion $p < 0.05$ is the same as a score above 12.706: “significant” in the thesis means a z-score above 12.706.

What it changes. In the Ukraine War corpus, 46 pairs occur more than twice:

Table C2. Pairs kept under each rule, Ukraine War corpus (46 pairs occurring more than twice).

Rule	Pairs kept
thesis: z-score > 12.706 (Table 6)	28
t-score > 12.706	0
t-score > 2	9

The two selections differ: four of the nine pairs kept by the t-score (*soldiers_killed*, *civilians_killed*, *people_injured*, *person_killed*) are not among the 28. In the Leipzig corpora the frequency floor already decides the result: all 127 pairs that occur more than 15 times in 2014 also pass under the t-score, so the choice of measure changes nothing there.

Table C3. Effect on each analysis that uses the association scores.

Where	Analysis	Effect
§4.2.2; Table 5; Figures 2, 19	the t-score as the most appropriate measure	<i>withdrawn</i>
§5.1.1; Tables 6, 7	the 28 significant Ukraine War pairs and their category proportions	the selection depends on the measure. Under both, civilian subjects and harmful verbs dominate (in the t-score selection, 86% and 95% of occurrences), so that observation holds. Restrictive verbs (<i>operations_stopped</i> , <i>units_shut</i> , <i>urey_captured</i> and four others) appear only in the z-score selection, and the percentages in Table 7 hold only for it
§5.1.1.2, §5.2.1.2; Tables 8–11, 15, 16	the filtered Leipzig pairs and their categories	unchanged for 2014 2023 scores not preserved; same frequency floor
§5.1.1.3; Figures 4, 5	category proportions compared across corpora	Leipzig side unchanged; Ukraine War side as for §5.1.1
§5.2.1.2, §5.2.1.3, §5.2.2.1; Tables 17, 18, 56	the Leipzig pairs as “more significant” than the Ukraine War pairs	<i>withdrawn</i> the p-values are not valid, and z-scores grow with corpus size, so averages from a 170,000-word corpus and corpora of 12 to 18 million words cannot be compared
§5.4.1; Tables 33, 34	filtered pairs in the Russian sources	read as for Table 6 (z-score above 12.706); not recomputed

The by-agent analyses (§5.3, and §5.4.2, which by its own account drops the p-value filter), the passive frequency series (§5.2.2) and the topic models do not use association scores and are unaffected. Throughout, the tables’ “t-score” columns are z-scores.

CHANGES A STATED FINDING

2. The decline in passive verb use (§5.2.2.1, Figure 8) does not hold

What the thesis says. Figure 8 and the surrounding text report that both passive sentences and passive verbs declined across 2014–2023, and read this as agreeing with Seoane & Loureiro-Porto (2005, cited in Di Ferrante 2023).

What is wrong. Table 18’s # Pass. V. column agrees with the passive-pair counts in Table 19 for 2016–2020, to within 1, but is misextracted for **2014, 2015 and 2023**, where it is off by 20,000 to 35,000:

Table C4. Passive-verb counts: Table 18 against Table 19, Leipzig corpora.

Year	Table 18 # Pass. V.	Table 19 # Pass. Pair	Difference
2014	86,172	120,683	–34,511 <i>misextracted</i>
2015	156,883	125,582	+31,301 <i>misextracted</i>
2016	123,474	123,474	0 matches
2017	126,258	126,257	+1 matches
2018	126,646	126,646	0 matches
2019	125,510	125,510	0 matches
2020	132,617	132,617	0 matches
2023	105,213	125,547	–20,334 <i>misextracted</i>

These three cells produce the downward slope. 2015 is too high early in the series and 2023 too low at its end; 2014, also too low, pulls the other way and only partly offsets them.

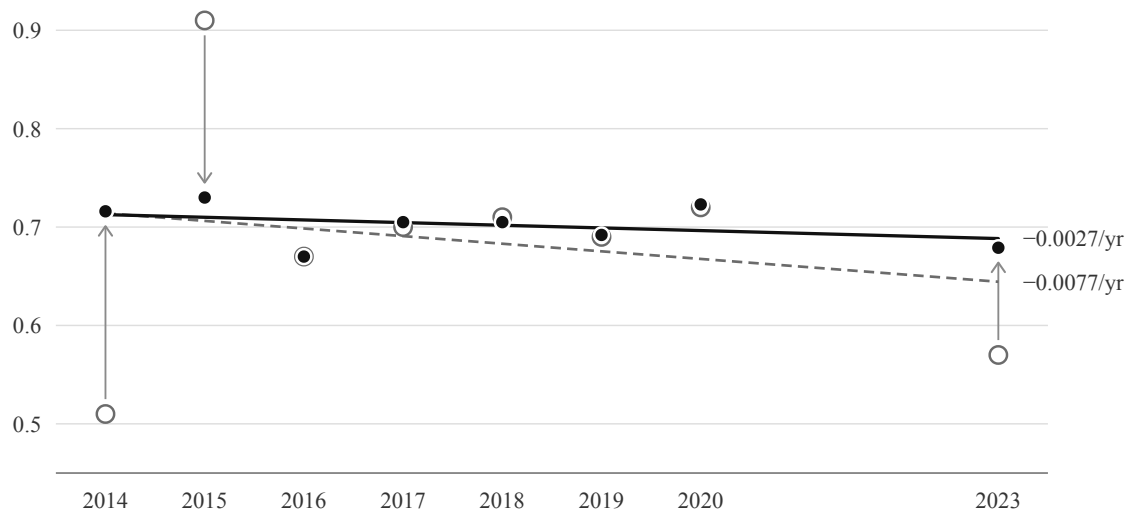
Corrected. Recomputing each year from Table 19’s pair counts over Table 2’s word counts gives:

Table C5. Passive verbs as a percentage of words, Leipzig corpora, as published and corrected.

	2014	2015	2016	2017	2018	2019	2020	2023
as published	0.51	0.91	0.67	0.70	0.71	0.69	0.72	0.57
corrected	0.716	0.730	0.670	0.705	0.705	0.692	0.723	0.679

The slope falls from –0.0077 to –0.0027 points per year. **The passive-verb series is flat, and the claim of a decline in passive verb use is withdrawn.**

Figure C1. Passive verbs as a share of all words, Leipzig news corpora, 2014–2023. Dots are yearly values; lines are least-squares trends.
–○– As published trend –0.0077 points/yr –●– Corrected trend –0.0027 points/yr



Arrows mark the three years whose published values were misextracted (2014, 2015, 2023). In the other years the two values differ by less than 0.01, so the dots overlap. There are no data for 2021–2022. Trends are fitted to the values in the table above.

What survives. The passive *sentence* series is unaffected — 18.69% in 2014 to 17.56% in 2023, slope –0.10/yr — and the decline reported there stands. So do the short-passive trend in §5.2.2.2, computed from Table 19, and the get-passive trend in §5.2.2.3, computed from Table 20.

Table C6. The diachronic trends in §5.2.2 after this correction.

Trend	Where	Status
Passive sentences, % of sentences	§5.2.2.1, Figure 8	Stands 18.69% (2014) → 17.56% (2023), –0.10/yr
Passive verbs, % of words	§5.2.2.1, Figure 8	<i>Withdrawn</i> series is flat, –0.0027/yr
Short passives	§5.2.2.2	Stands
Get-passives	§5.2.2.3	Stands

3. Table 18’s %Pass. v. column uses two denominators

§5.2.2.1 states that passive verbs are expressed as a percentage of total **words**. That holds for the Leipzig rows (0.51 to 0.91, average 0.69), but the Ukraine War row divides by total **verbs**, active plus passive: 1,473 / (8,909 + 1,473) = 14.19%. The column therefore sets 0.69 beside 14.19. On the stated basis the Ukraine War figure is 0.86% of words.

The text compares a Leipzig average of 12.90% with the Ukraine War’s 14.19%. Both are on the verb denominator, so that comparison is like with like, but 12.90% does not appear in Table 18, whose Leipzig average is 0.69%. The comparison with Di Ferrante (2023) that follows uses the word denominator and is unaffected.

Table C7. Passive verbs on each denominator.

	÷ total words (% of words)	÷ total verbs (% of verbs)
Ukraine War	0.86	14.19
Leipzig average	0.69	12.90

Table 18 prints the two outlined figures in one column, although they sit on different denominators. Reading down a column here compares like with like. 0.86 is recomputed; 12.90 is printed in the text only.

4. Table 25 contains a duplicated row

citizen_detained · by_pro_-_Russian_separatists_forces is printed twice — 21 rows, 20 distinct. Table 26 is clean at 16. The §5.3.1 percentages use 21, so on distinct rows: restrictive $10/20 = 50.00\%$ (published 52.38%) and violent $3/20 = 15.00\%$ (published 14.29%). The subset total is 36, not 37. The finding — violent predicates concentrating on non-human agents, restrictive on human agents — is unaffected.

Table C8. §5.3.1 figures with the duplicated row removed.

	Published (21 rows)	Distinct rows (20)
Human agents, restrictive	11 · 52.38%	10 · 50.00%
Human agents, violent	3 · 14.29%	3 · 15.00%
Subset total, Tables 25 + 26	37	36

5. Russia/Russian as agent: the schema is computed on 35 of 37 clauses (§5.3.2)

Table 29 prints 37 clauses. The schema beneath it counts 35 (14 restrictive + 12 violent + 4 negative + 3 neutral + 2 positive), and all five percentages — and the 85.72% in the interpretation — are computed on 35. Two clauses are uncategorised.

The corresponding Ukraine figures are correct: $10+9+8+3+2 = 32$ matches Table 32, and 56.25% / 43.75% follow. Russia-as-modifier is also correct at 11 clauses and 72.72%.

Figure C2. Clauses printed against clauses counted in the schema.



Table C9. Russia/Russian as agent: the schema's percentages and their denominator.

Verb class	Clauses	Published % (of 35)	Share of all 37 printed clauses
Restrictive	14	40.00	37.84
Violent	12	34.29	32.43
Negative	4	11.43	10.81
Neutral	3	8.57	8.11
Positive	2	5.71	5.41
Uncategorised	2	—	5.41
Restrictive + violent + negative	30	85.72	81.08

6. “Leipzig 2014/2023” names two different preparations

§3.1.2 describes three cleaning steps and reports that they leave “12 and 13 million words” — but points to Table 2, which shows 16.9 and 18.5 million.

Table 2 reflects the first two steps only. Table 14 (570,512 sentences / 12,322,647 words for 2014; 625,932 / 13,224,845 for 2023) reflects all three, including the removal of lines containing direct speech. Sentences present in Table 2 but not Table 14 average 24.4 words against 21.6 for those kept in 2014, and 23.0 against 21.1 in 2023, consistent with quote-bearing news lines being longer.

§5.2.1 (the pair-level lookup) uses the fully cleaned version; §5.2.2 (Tables 18–20, the eight-year series) uses the partially cleaned one. Both are internally consistent; the error is that §3.1.2 describes one and cites the other.

Table C10. Two preparations of the same Leipzig corpora.

	Table 2	Table 14
Cleaning steps	1. line numbers removed 2. lines with fewer than 10 words and more than 4 numerical digits removed	Steps 1 and 2, plus 3. lines with a double quotation mark (direct speech) removed
Leipzig 2014	756,346 sentences 16,859,714 words	570,512 sentences 12,322,647 words
Leipzig 2023	854,997 sentences 18,498,430 words	625,932 sentences 13,224,845 words
Used in	§5.2.2 (Tables 18–20, the eight-year series)	§5.2.1 (the pair-level lookup)
In §3.1.2	cited	described (“12 and 13 million words”)

Figure C3. Words removed by step 3 (direct speech).



Table C11. Sentence length of the lines removed by step 3.

Year	Sentences removed	Words per sentence, removed	Words per sentence, kept
2014	185,834 (24.6%)	24.4	21.6
2023	229,065 (26.8%)	23.0	21.1

7. “Passive pair” is used for two different quantities

Table 14’s pair counts (Ukraine War 1,061; Leipzig 2014 60,973; Leipzig 2023 64,688) are **distinct pairs**. Table 19’s (Leipzig 2014 120,683; Leipzig 2023 125,547) are **occurrences**. The text calls both “passive pairs”. The two tables also count different preparations of the Leipzig corpora (correction 6), so they cannot be compared row by row.

For the Ukraine War corpus, the pair-frequency table has 1,061 rows whose frequencies sum to 1,330; it is available from the author on request.

Table C12. “Passive pair” in two tables, Leipzig 2014 and 2023.

Corpus	Table 14 distinct pairs	Table 19 occurrences
Leipzig 2014	60,973	120,683
Leipzig 2023	64,688	125,547

8. Table 4’s expected frequencies belong to another pair (§4.2.1, §4.2.2)

Table 4, the contingency table for *sovereignty_threatened*, gives expected frequencies of 94.511 and 1,162.489 for the two cells where the subject is absent. Those are the values for *people_killed*. From the table’s own margins they are 4.985 and 1,321.015; as printed, the four expected frequencies sum to 1,261 rather than 1,330. The chi-squared worked example in §4.2.2 uses 1,162.489 and so gives 1,211.03; with the correct value it is about 1,063. The scores in Table 6 match a recomputation from the released counts, so only the printed example is affected.

Table C13. Table 4, subject-absent row: printed and correct expected frequencies.

	Printed	Correct
Verb present	94.511	4.985
Verb absent	1,162.489	1,321.015
Sum of all four expected frequencies	1,261	1,330
Chi-squared, worked example in §4.2.2	1,211.03	≈ 1,063

9. Two Average rows are miscalculated

Table C14. Average rows in Tables 2 and 33, published and correct.

	published	correct	difference
Table 2, mean sentences	826,347	823,249	−3,098
Table 2, mean words	17,829,303	17,914,504	+85,201
Table 2, mean words/sentence	21.82	21.78	−0.04
Table 33, mean words	210,737	210,070	−667

Each correct value is the mean of the column above it, as in the thesis’s other Average rows; for words per sentence, total words over total sentences gives 21.76. Tables 18–20 print the correct Leipzig averages; only the two dataset tables are wrong.

10. Abstract corpus size

The abstract gives the Leipzig corpora as “17M”. This is the eight-year average on Table 2’s basis and reads against §3.1.2’s “12 and 13 million” — see correction 6.

Table C15. Two corpus sizes, two bases.

Where	Printed	Basis
Abstract	“Leipzig news corpora (17M)”	Table 2 preparation (steps 1–2), eight-year average
§3.1.2	“12 and 13 million words”	Table 14 preparation (steps 1–3), 2014 and 2023

11. The links in Appendices 1 and 2 are dead

Appendix 1 (corpora) and Appendix 2 (source code) link to files at github.com/aminomewza/Who-was-done-what-master-s_thesis–, each pinned to a specific commit. The repository is now at https://github.com/n-sanitdee/Who-was-done-what-master-s_thesis–, and the pinned commits are no longer in it, so the old links return 404 even with the new account name.